

Lab Assignment 3

Speech Emotion Recognition Systems

Spring 2026

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1 Logistics

- Group Size: 4 students
- Total Points: 10 points
- Length: Maximum 8 pages (double-spaced, 12pt font)
- Submission: Each group member must individually submit the same final copy to Canvas (the exact document your group developed together). Make sure all group members' names are listed on the report.

Submission: Submit one PDF to Canvas. Include your name at the top of the report.

Code Requirement: You must also submit:

- Your full code
- Instructions to reproduce results
- Any dataset split information

Formatting requirement: You must organize your report using the exact numbering and section headings provided below (1–8). Clearly label each answer under its corresponding section.

2 Objective

The goal of this lab is to improve the robustness of a Speech Emotion Recognition (SER) system.

In the previous lab, we used a pre-trained wav2vec2-based emotion model. However, the model was trained on limited data and may not generalize well to new speakers or datasets.

In this assignment, you will design and evaluate a more reliable audio classification pipeline.

Learning Goals:

By completing this lab, you should understand:

- Limitations of small fine-tuned models
- Dataset size and generalization effects
- Differences between deep learning and classical ML approaches
- How to evaluate model robustness critically

3 Task Overview

You must choose **ONE** of the following approaches:

Option 1: Use a Stronger Pre-Trained SER Model

- Research and select a different Hugging Face SER model.
- Justify your choice.
- Test it on RAVDESS (or another dataset).
- Compare performance with the original model.

Option 2: Fine-Tune wav2vec2 on RAVDESS

- Assemble the RAVDESS/Toronto speech dataset.
- Fine-tune wav2vec2 for emotion classification.
- Evaluate performance.
- Compare before vs. after fine-tuning.

Option 3: Classical Baseline (Traditional Classifier)

- Extract features.
- Train a classifier (e.g., SVM, Random Forest).
- Evaluate performance.
- Compare with deep learning performance.

Note 1: Plan and manage your group's timeline and task distribution carefully. Clearly assign responsibilities to each team member to ensure balanced contribution. A suggested high-level timeline might look like this:

Week 1:

- Prepare dataset
- Implement the training pipeline for the chosen Option
- Run baseline and initial experiment

Week 2:

- Refine model (limited adjustments)
- Evaluate results (metrics + confusion matrix)
- Perform error analysis
- Write and finalize the report

Note 2: For all options, the focus is on analysis and methodological understanding, not achieving state-of-the-art accuracy. Focus on steady progress each week rather than leaving experimentation and analysis to the final days.

4 Dataset Description

- What dataset did you use?
- Number of samples per emotion class
- Train / validation / test split
- Any preprocessing performed (resampling, silence trimming, normalization)

5 Methodology

5.1 Feature Extraction

- What features were used?
 - Raw waveform
 - wav2vec embeddings
 - MFCC
 - Other
- Why did you choose this method?

5.2 Model Architecture

- Describe the model used:
 - Pre-trained transformer?
 - Fine-tuned model?
 - Classical ML model?
- Key hyperparameters (epochs, learning rate, batch size, etc.)

6 Evaluation

6.1 Quantitative Metrics

- You must evaluate your model using quantitative metrics. Required Metrics include:
 - Accuracy
 - Confusion matrix
 - (Optional) Precision, Recall, F1-score

6.2 Baseline Comparison

You must compare your model against the original baseline wav2vec2 model used in the class.

- Present results in a comparison table:

Model	Accuracy	Average Confidence	Notes
Baseline Model			
Your Model			

- Briefly describe:
 - Did performance improve?
 - Did confidence values increase?
 - Were predictions more stable?

6.3 Performance Analysis

Discuss:

- Which emotions were easiest to classify?
- Which emotions were most frequently confused?

- Were certain speakers harder to classify?
- How confident was the model overall?

7 Results & Analysis

Provide a detailed analysis of your results.

7.1 Error Analysis

Select at least **three misclassified samples** and:

- Show the true label vs the predicted label
- Explain why the model may have made this mistake
- Identify any confusion patterns (e.g., happy vs surprised, calm vs neutral)

7.2 Model Behavior & Robustness

Discuss:

- Did your approach improve robustness compared to the baseline?
- Did fine-tuning reduce random-like predictions?
- Did classical ML behave differently from deep learning?
- How sensitive was your model to speaker variation?

7.3 Interpretation of Results

Explain possible reasons for your observed performance:

- Dataset size limitations
- Model capacity

- Overfitting or underfitting
- Domain mismatch
- Preprocessing effects

8 Limitations & Future Improvements

8.1 Identified Limitations

Clearly describe **at least three specific limitations** of your approach. These may include:

- Limited dataset size
- Class imbalance
- Speaker variability
- Overfitting or underfitting
- Domain mismatch (training vs test data)
- Model capacity limitations
- Hardware or training constraints

For each limitation, briefly explain:

- Why does it affect performance
- How it appeared in your results

8.2 Generalization Concerns

Discuss whether your model would generalize well to:

- New speakers
- Different accents

- Different recording environments
- Real-world audio

Support your answer using evidence from your results.

8.3 Future Improvements

Propose **at least two realistic improvements**, such as:

- Using more data
- Data augmentation
- Better hyperparameter tuning
- Cross-validation
- Alternative model architectures
- Transfer learning from larger datasets

Briefly explain how each improvement could increase robustness or performance.